The effects of mergers and acquisitions on firm performance

An international comparison between the non-crisis and crisis period

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Abstract

This paper investigates the effects of mergers and acquisitions (M&As) around the world. It presents evidence that M&A announcements have positive effects on stock prices. It also finds that the wealth effect is larger for target shareholders compared to acquirer shareholders and that the effects in a crisis period outperform the effects in a non-crisis period. Looking at post-merger profitability, M&As are profitable in at least the first five post-merger years, where the results in the non-crisis period outperform the results in the crisis period. This means that stock prices react more positively during a crisis compared to a non-crisis period, while for post-merger profitability an opposite effect is found. The effects across countries look similar, indicating that firms in different countries show, in general, the same effects, both in terms of wealth effects and post-merger profitability, when an M&A occurs.

Table of Contents

1. Introduction

"Sometimes your best investments are the ones you don't make" (Trump, 1987) is a quote of business magnate Donald Trump. It refers to the question whether it is wise to invest in a new project or not. A new investment means often the upside potential of high profits, but on the other hand there is the risk of making a bad investment decision and losing a lot of money. In the ideal world an investor knows the outcome of his investment in advance. However this is not realistic and therefore an investor will always face risk on his investments. But wouldn't it be great when at least the general effects of an investment were clear in advance? Mergers and acquisitions (from now on M&As) are common types of investments with in most cases unexpected outcomes. Knowing in advance the general effects of M&As in a certain country and/or a specific economic situation would make it easier to make M&A investment decisions.

This thesis investigates the effects of M&As on firm performance. In the current literature the effects of M&As are widely discussed, but strikingly, most of those papers are relative old (written begin 2000s or earlier). Therefore they use databases before the sixth merger wave (2003-2007) and thus also before the economic crisis started in 2008. This crisis had a large influence on the world economy and therefore probably also on firm behavior regarding M&As. It seems obvious that firms tend to make different M&A investment decisions during a crisis period. However, whether M&A effects during a financial crisis differ from the effects in a non-crisis period is not clear. Despite several studies of M&A effects in different countries, there is also no clear answer on the question whether the M&A effects differ across countries. Most international studies so far focused on the worldwide effects of M&As or just on the effects in a specific country or region. Gugler, Mueller, Yurtoglu and Zulehner (2002) are an exception and are one of the few researchers who made an international comparison to expose the different M&A effects in countries all over the world. However, also in their study no decisive answer has been found.

In this study the M&A effects are investigated from several perspectives. First of all, this study uses a recent dataset (2002-2013) to be able to re-examine older studies and finds the current effects of M&As on firm performance. In the second place, this study focuses on international differences. This makes it also possible to re-examine international studies like Gugler et al. (2002) and to find an answer on the question whether M&A effects differ across countries. For example, do U.S. mergers have the same effects on firm performance as European mergers? In the third place, this study focuses on the influence of the economic crisis on the effects of M&As. Therefore it presents an international analysis of the influence the financial crisis, which started in 2008, had on firm performance when an M&A took place. For this, a comparison between the pre-crisis period (2002-2007) and the crisis period (2008-2013) is made. This approach gives the possibility to answer questions as: Do merged companies have an advantage compared to their competitors during an economic crisis? And is it more valuable to execute an M&A during a crisis period or during a non-crisis period?

The most existing studies on the effects of M&As on firm performance can be divided into two groups. The first group studied the effects on stock price changes after an M&A announcement, while the second group studied the effects on profitability for the merged firms. To be able to measure both effects and to re-examine the previous literature, two different approaches are used in this study. The first approach includes an event study and a cross-sectional regression to measure and explain the stock price reaction when a firm announces an M&A. The second approach follows the method developed by Gugler et al.

(2002) and measures the effect on firm performance by the change in profits (EBIT). These two approaches are applied for different country groups in a non-crisis and a crisis period. This makes it possible to show the influence of the crisis on the effects of M&As in several countries. Summarizing this, the following main research question can be formulated:

Are the effects of M&As on firm performance positive, both before and during the economic crisis, when investigating a worldwide sample of M&As between 2002-2013?

This research question is investigated from two approaches:

Approach 1: Firm performance measured by stock prices.

Approach 2: Firm performance measured by profits.

The results from approach 1 show that M&A announcements have, in most cases, a positive effect on stock prices, where the results for target shareholders are more positive than for acquirer shareholders. In addition to this result, evidence is found that M&A announcements in a crisis period result in higher stock returns than M&A announcements in a non-crisis period. The second approach shows that M&As in general are profitable in at least the first five post-merger years. For this approach the most positive results are found in the non-crisis period, which contradicts the results of the event study. Regarding the differences between countries, this study finds that the effects are similar. This means that all country groups show mostly the same effects. However, the magnitude of the effects differ, but are not structural larger or smaller for certain countries.

This paper is organized as follows. Section 2 covers the literature review. Section 3 presents the data and methodology used to answer the main research question. Section 4 contains the results and finally sections 5 and 6 present the conclusions and limitations of this study together with some recommendations for future research.

2. Literature review

M&As are very common in the financial environment. Hence, it is no surprise that since the first merger wave many academic papers were written about this topic. The first wave started at the end of the 19th century and was followed by five other merger waves at the end of the 1920s, 1960s, 1980s, 1990s and the mid-2000s. This literature review covers the following subjects: causes of M&A activity, effects of M&As, M&As in different countries, M&As in different economic periods and determinants of abnormal returns, which will be touched upon briefly in the next sections.

2.1. Causes of M&A activity

In the literature different clarifications can be found concerning the causes of M&As. Trautwein (1990) studied various theories regarding merger explanations and classified them into seven groups. Subsequently these seven groups can be ordered into three groups with a different level of plausibility. The first group has the highest plausibility and consists of the Valuation, the Empire-building and the Process theory. The second group consists of the Efficiency and the Monopoly theory, but is less plausible than the first group. The third group consists of the Raider and the Disturbance theories, but is rather implausible. The Valuation theory argues that acquirers have better information about a target firm than the market and can therefore make better estimations. This makes it possible to benefit from undervalued firms (Holderness and Sheehan, 1985). According to the Empire-building theory, mergers are driven by managers who maximize their own utility instead of shareholders' value (Trautwein

1990). The Process theory claims that the decision to merge is driven by the strategic decision process of a firm. However, in practice it is realistic that a decision process is not entirely based on rational choices, it is also influenced by political factors, organizational routines, limited information accessibility and agency issues (Trautwein, 1990 and Herbert, 1957). The Efficiency theory explains mergers as being planned and executed to achieve synergies. Three types of synergies can be distinguished. Financial synergies which result in lower cost of capital, operational synergies where firms combine their operations to lower costs and/or to offer unique products/services and managerial synergies where the managers of the acquiring firm improve the planning and monitoring of the target firm (Trautwein, 1990). The last relevant theory in Trautwein's (1990) study is the Monopoly Theory. This theory views mergers as a way to achieve market power with a horizontal or vertical M&A.

The study of Trautwein (1990) shows the general motives to merge, but looking into more detail, underlying causes can be found. Healy, Palepu and Ruback (1992) studied the influence of M&As on firm performance and found that merged firms show significant improvements in asset productivity relative to their industries, leading to higher operating cash flow returns. These improvements are primarily explained by the book values of asset disposals. Before the merger this value was 0.9% and after the merger it was 1.3%. Looking at the numbers of their industry counterparts, they found values of 0.1% and 0.6% respectively. Andrade, Mitchell and Stafford (2001), endorse the conclusions of multiple event studies and conclude the following: "mergers create value for stockholders of the combined firms, with the majority of gains accruing to the stockholders of the target". This statement suggests that in general M&As are profitable and have a positive impact on the stockholders of a merged firm. The effects of M&As found by Healy et al. (1992) and Andrade et al. (2001) show that improving profitability and maximizing shareholder wealth are main causes to merge. Comparing these findings with the theories of Trautwein (1990), these results are aligned with the Efficiency and Valuation theory. The findings of Healy et al. (1992) can be explained by the Efficiency theory, due to the improvement in productivity. The results of Andrade et al. (2001) can be explained by the Valuation theory due to the realized increase in shareholder value.

The theories formulated by Trautwein (1990) explain the motives to merge from a firms' internal perspective. However, the literature shows that those internal reasons could be encouraged by external influences, like the importance of the industry structure and the economic environment. A merger can for example be caused by technological and supply shocks, which possibly result in excess production capacity in many industries (Jensen, 1993). M&As could be a solution to this overcapacity because expanding makes it possible to prevent shrinkage and to continue the use of full production facilities. Besides the technological and supply shock, other major shocks can cause a merger, such as deregulation, increased foreign competition, financial innovations and oil price shocks (Mitchell and Mulherin, 1996 and Harford, 2005). A study of the most recent merger wave showed that in this merger wave the low financing rates and companies' rich cash balances played a large role in the M&A decision (Alexandridis, Mavrovitis and Travlos, 2012).

2.2. Effects of M&As

According to the formulated motives to merge, it can be expected that the execution of a merger or acquisition is a well-considered decision, with expected benefits accruing to the firms involved as a result. The results of Healy et al. (1992) confirm this expectation and show positive effects on the profitability of the merged companies. The hypothesis is also underlined by Gugler et al. (2002), who conclude that mergers in general result in significant profit increases for the combined company. Nevertheless, there are also studies, such as the study of Ravenscraft and Scherer (1989), who found a decline in the profitability of acquired firms. The reason for this decline is losing the control due to more complex organizational structures and lessened managerial competence and/or motivation. With this explanation Ravenscraft and Scherer (1989) show considerable skepticism towards the claim that mergers are on an average efficiency-enhancing. The effect of a merger or acquisition could have a negative influence on the firms' industry and facilitate industry contraction. Own-industry M&As had a negative relation to capacity utilization in the 1980s and therefore induced exits in industries with excess capacity (Andrade and Stafford, 2004). Then again, Andrade and Stafford (2004) add to this conclusion that mergers can play two different roles and therefore show positive effects on capacity utilization in the 1990s (which is also confirmed by Jensen, 1993). It therefore can be concluded that mergers do not automatically result in better firm performance. M&As could negatively affect the merged companies or the entire industry. Based upon the numerous studies that deal with M&A activity and its effects, it is not possible to give a decisive answer on the question whether mergers have a negative or positive effect on the profitability of firms.

Besides the post-merger effect it is also important, in case of determining the effects of M&As on firm performance, to take into account the stock market reaction around the M&A announcement dates. This reaction reflects the changes in expected future cash-flows that will accrue to the shareholders of the firms involved and can be seen as a proxy of the expected value arising from the merger (Campa and Hernando, 2004). Looking at the separate effects for the bidder and target firms around the M&A announcement date, a trend can be found in the literature. Dodd and Ruback (1977) conclude that in the month of the M&A announcement, the stockholders of the target firm earn large and significant abnormal returns of 20.58%. For the stockholders of the acquiring firm, they found a lower, but also positive, return of 2.83%. Also Franks and Harris (1989) conclude in their study that target firms benefit from a merger or acquisition around the announcement date, but regarding the acquiring companies they conclude that those companies earn zero return or generate just a modest gain. Andrade et al. (2001) move even further and found, besides the positive effect for the combined firm, negative abnormal returns for the acquiring firms around the announcement date. In a more recent study of Netter, Stegemoller and Wintoki (2011) the trend in returns was underlined. They conclude that the acquiring firms gain an abnormal return of 1.1%, while the target firms gain an abnormal return of 20.4%. To explain this trend, different causes can be found. Netter et al. (2011) conclude in their study that there is not an obvious reason to explain the trend. This is confirmed by other studies which conclude that several factors play a role. Andrade et al. (2001) suggest that M&As are not always undertaken for the good reasons. For example empire-building by managers can favor a merger, which is not always beneficial for the acquirer's shareholders. Another reason can be the presence of competing bidders, which results in higher takeover prices. Other possible explanations, found by Moeller and Schlingemann (2004), are deal atmosphere (friendly takeovers result in better results for acquiring firms than hostile takeovers) and the size of the acquirer (the larger the acquiring firm, the better the acquirer's results).

A slightly deviating trend is found by Moeller, Schlingemann and Stulz (2005). They found that the profitability of M&As depends on its timing. Acquisition announcements in the 1990s were profitable until 1997, but in the subsequent years until 2001 there were losses, which wiped out all the gains made earlier. However, in this study, the negative outcomes are caused by a relative small number of acquisitions with a negative effect which are weighted more heavily than the positive effects of thousands of other acquisitions. Therefore, regarding the effect of M&A announcements a clear trend, where the target's abnormal returns are higher than the acquirer's abnormal returns, is observable. However, regarding the abnormal returns of the acquiring firms, there is no clear answer whether these returns are positive or negative. Regarding the returns of the target firms, it can be concluded from several studies that those abnormal returns are in general positive, especially due to the improvement in efficiency, control, economies of scale, synergies and the advantage of certain deal characteristics (for example deal attitude and payment method).

The effects of M&As can also be studied per industry. This is meaningful because M&As could have different effects in different industries. For example the study of DeYoung, Evanoff and Molyneux (2009) who tracked the evolution of M&A literature concerning financial institutions in the post-2000 period. In the pre-2000 literature most studies concluded that bank mergers were efficiency improving, while event-study literature found no convincing evidence for positive stockholder wealth effects. This result motivated researchers to find other explanations for the efficiency improvement. DeYoung et al. (2009) reviewed more than 150 studies in the post-2000 period and concluded that U.S. bank mergers are in general efficiency improving, although there is still no convincing evidence for positive stockholder wealth effects in the financial sector. In contradiction with this result, in European bank deals they found a clear consensus, which shows both efficiency gains and enhanced stockholder value. Regarding the U.S. deals, most studies found a positive relationship between CEO compensation and merger activity. The overall findings in the study of DeYoung et al. (2009) indicate that M&As in the financial sector have a positive effect on firm performance, but they also warn for the potential negative impact on prices for consumers, credit availability, too big to fail issues and market power effects.

For the pharma-biotech industry, Danzon, Epstein and Nicholson (2007) found that mergers in this industry are mostly undertaken as a solution to solve disappointing profits. However, they found no evidence that mergers create positive long term values, which suggests that M&As do not result in cost savings and economics of scale in this industry. Fowler and Schmidt (1988) studied the influence of M&As in the manufacturing industry. They conclude in their study that M&As do not affect financial performance in a favourable manner. Their results indicate that accounting and investor returns decrease significantly in the four years after the M&A compared with the four years before the M&A. A study to M&As in different Indian industries (Martynova and Renneboog, 2008) illustrates the influence that an industry could have on the M&A effect. This study found a positive impact on profitability for firms in the banking and finance industry, but a negative impact for firms in the pharmaceutical, textile, chemical and agribusiness industry.

2.3. M&As in different countries

The effects of M&As discussed so far were not specified per country and therefore illustrated a general view without considering the geographical influences. The fact that geographical closeness plays an important role in the M&A discussion is investigated by Böckerman and Lehto (2006). They studied domestic M&As from a regional perspective and found that a great number of M&As occur within defined regions. Their results show that companies prefer to seek partners from their home regions, that larger companies overcome geographical boundaries in general more easily and that domestic M&As are more likely to occur in regions that contain a great number of companies. Regarding the international effects of M&As several papers are written. Mueller (1997) summarized the results from 20 studies drawn from 10 countries regarding the post-World War II period. His first finding shows that the most common way to measure the effects of mergers on profitability is by computing the weighted average profit rate of the merging firms before the merger and compare it to the profit rate of the merging firms after the merger. To control for economic conditions the change in profits of the merged firms should be compared with a control group of similar non-merging firms or industry averages of the merging firms.

According to the study of Mueller (1997), the M&A effects in the United States correspond with the effects in the United Kingdom. In those countries most evidence points toward no increase and probably some decline in the profitability of the merging firms after the merger. However, the international results are mixed because some other studies found for both countries the opposite effect (Cosh, Hughes and Singh, 1980). Mueller (1997) concludes for other countries that mostly no distinct patterns could be found and that most results are statistically insignificant. In some countries mergers seem to result in profit increases, while in other countries they seem to result in profit decreases. Alexandridis, Petmezas and Travlos (2010) have made also an international comparison and provide evidence that public acquisitions do generate gains, but the distribution of gains between acquiring and target firms depends on the degree of competition in the market. Acquirers in the most competitive takeover markets (U.S., U.K. and Canada) realize higher gains than acquirers in other countries. Regarding abnormal returns they found in most countries the same general view as discussed earlier in this study, where the acquirer's abnormal return is lower than the target's abnormal return.

Other international results regarding abnormal returns were found by Park (2004). He found significant greater abnormal returns for US firms than non-US firms, which indicates that the U.S. stock exchange responds more positively on M&A announcements than other stock exchanges. Campa and Hernando (2004) focused in their study on shareholder value in

European M&As. The process of economic integration and the deregulation of economic activities in the European Union (E.U.) were important developments in the late 1990s and begin 2000s. Those developments stimulated the restructuring of companies in the E.U. and especially the restructuring of companies located in the euro area. The M&A patterns found in this period are the same for European M&As as for M&As in other countries. This means that target shareholders received in general positive cumulative abnormal returns (9%) and that acquiring firms received no cumulative abnormal return (0%). Looking at the geographical dimension of merger deals in Europe, it is important to take into account institutional and policy barriers. M&As in countries with more regulatory frameworks generate lower cumulative abnormal returns than M&As in less regulated countries. The differences range between 1.3% and 5.2% for target firms and between 1% and 3.5% for acquiring firms.

Gugler et al. (2002) performed an international comparison focused on post-merger effects. They found that globally 56.7% of all M&As result in higher than projected profits and that almost the same fraction of M&As result in lower than projected sales after five years. Therefore, using profits as a measure for success gives the opposite result than using sales as a measure. From this result Gugler et al. (2002) conclude that M&As on average do result in significant increases in profits, but on the other hand reduce the sales of the merging firms. In the international comparison, Gugler et al. (2002) found that post-merger patterns of profits and sales changes look similar across countries. Though they found mixed results in significance levels, in the U.S., U.K. and Europe patterns which indicate a positive effect on profit and a negative effect on sales were found. M&As in Japan, Australia, New Zealand and Canada seem to have the opposite effect on profit, but this outcome is probably affected by their small sample sizes.

Looking at some country specific studies, significant patterns of profit increases can be seen in Canada (Baldwin, 1995) and Japan (Doi and Ikeda, 1983) and profit decreases in the Netherlands (Peer, 1980) and Sweden (Edberg and Ryden, 1980). A study of M&As in the Indian Oil and Gas sector (Desai, Joshi and Trivedi, 2013), finds evidence that M&As do not create immediate shareholder wealth and profit margins for the acquiring firm. However, in the long term a merged company would be able to better cope with competition, decrease costs and realize grow in continuously changing business environments.

The literature shows also that country features play a role as determinants of M&As. Rossi, Paolo and Volpin (2004) studied firms in 49 different countries and showed that laws and enforcement are important determinants. Differences in laws and enforcement over countries explain the intensity and the patterns of M&As around the world. This finding confirms the conclusion of La Porta, Lopez-de-Silanes, Shleifer and Vishny (1997) who argued that legal environments differ across countries and that these differences are important for financial markets. In countries with better accounting standards and strong shareholder protection the M&A market therefore is more active. In line with this conclusion, Rossi et al. (2004) showed that firms in countries with weak investor protection are often sold to buyers from countries with strong investor protection. Another result that has been found in this study is that in countries with better shareholder protection hostile takeovers are relatively more likely. A one-point increase in shareholder protection, results in 0.8 percentage point more hostile takeovers. This is primarily due to the fact that shareholder protection makes control more debatable by reducing the private benefits of control. Other results found in the study of Rossi et al. (2004) are higher takeover premiums in countries with higher shareholder protection and less all-cash bids in countries with more shareholder protection for the acquirer. This last statement indicates that M&As paid with shares need an environment with more shareholder protection.

Thus, the results from international studies regarding M&A effects are mixed. It can be concluded that economic integration, deregulation of economic activities and country features play a role in the M&A process. Regarding the effects of M&As on firm performance, the literature shows that country specific results are less significant than results from studies with broader samples. Already been exposed in this literature review (by the conflicting conclusions regarding post-merger effects found in previous studies), was the difficulty of finding a decisive answer on the question whether M&As have a positive or negative influence on firm performance. For example the conclusions of Gugler et al. (2002), who showed a positive effect on firm performance and Ravenscraft and Scherer (1989), who showed a negative effect on firm performance. It can be concluded that finding an international link and exposing clear effects of M&As across countries is even more challenging.

2.4. M&As in different economic periods

This study is focused on the 2002-2013 period. This means that it includes years during the sixth merger wave and years during the global economic crisis. The sixth merger wave started in 2003 and came to an end in 2007. According to Harford (2005), a number of aspects are needed to start a merger wave. In the first place, merger waves need a cause, which is in most cases an economic, regulatory or technological shock. Thereby access to sufficient capital liquidity is essential to propagate a wave. This is also the reason why merger waves arise often after a financial recession. At that moment a rapid expansion of credit, caused by growing capital markets and growing stock markets, is often observable. Therefore waves occur mostly in a positive economic and political environment. However, Rhodes-Kroppf and Viswanahtan (2004) show that those explanations are not the full story. They studied the misvaluation of the market and elaborate the idea that even fully rational participants make mistakes. They found that in case of an overvalued market, the target rationally values the offer correctly and therefore reduces the expected value of a given stock offer. However, the target is still more likely to overvalue the offer when the market overvaluation is greater, although the fact that its own stock is also affected by the same market overvaluation. This means that overvaluation increases the chance that a merger occurs. So, an overvalued market could start a merger wave even when there are no other reasons. Viewed from the other side, waves could be stopped by an undervaluation of the market.

The assumption made in the literature that mergers are driven by managers timing the market is not supported by Harford (2005). This assumption has certainly influence on some mergers, but it cannot be seen as the cause of merger waves (Harford, 2005). However, due to overoptimistic managers it can be said that M&As towards the end of a merger wave become progressively more driven by non-rational managerial decision making (Martynova and Renneboog, 2008). Further Martynova and Renneboog (2008) showed that merger waves are usually disrupted by a sharp decline in stock markets and a period of economic recession. Most of the characteristics described are also recognizable regarding the sixth merger wave. This merger wave started after the crash of the dot-com bubble, a lot of credit (against low financing rates) was available, there was a positive economic environment and it was followed by a period of economic recession.

The second part of the database used in this study contains the years after the sixth merger wave (2008-2013) and hence includes the period of the global economic crisis. This crisis, developed with remarkable speed, started in the late summer of 2008 (Kotz, 2009). Reason for the start of the crisis was the collapse in mortgage-related securities that had been spread through the U.S. and the global financial system. Due to the international spread, a major part of the world's financial system was damaged. After this global crisis the European sovereign debt crisis started. Some European countries had problems with repaying their debts and the yield spread, an indicator of stress in the market, between a country's bonds and those of Germany climbed enormously (Shambaugh, Reis and Rey, 2012). In 2009 the world recovered from the global recession and growth started again. However, Euro-countries had problems with recovery and fell behind compared to countries like the U.S., U.K. and Japan.

Regarding M&A activity during the global economic crisis, Gaughan (2009) found that the volume of M&A activity decreased. Credit became increasingly harder accessible, which caused an increase in the costs of financing M&As. In addition, sales and profits during a crisis often decline, which makes it unattractive to accomplish a merger or acquisition. Moreover, Ang and Mauck (2011) studied the effects of fire sales in the M&A market during an economic crisis. They found out that acquiring firms do not benefit from acquiring distressed companies, due to the misconception regarding the benefits of fire sales discount which in reality does not exist. It can be concluded that acquiring other firms during an economic crisis is difficult and unattractive and as a consequence less M&As occur during a financial crisis.

2.5. Determinants of abnormal returns

The effect of M&As on shareholder value can be determined by calculating and analyzing abnormal returns in an event study. But which characteristics have influence on the occurrence of abnormal returns? Many potential determinants, which can be divided into deal characteristics and firm characteristics, are described in the literature. In this section the most common determinants in literature will be discussed.

Braggion, Dwarkasing and Moore (2012) did a cross-sectional analysis of acquirer's and target's abnormal returns. Although their study just focused on the banking sector in the U.K., they found that the lower the target's ROE, the higher the acquirer's abnormal return. This reflects the possibility to improve the profitability of a poorly performing target firm by the management of the acquirer. Regarding the abnormal returns of the target firms, some weak effects were indicating that less profitable targets experience larger abnormal returns. Braggion et al. (2012) also added control variables for various measures of ownership concentration, but these variables did not seem to be very important.

Another relevant firm characteristic was found by Schleifer and Vishny (2003). They investigated stock market acquisitions and found out that firms with overvalued equity are able to make acquisitions, while firms with undervalued or relatively less overvalued equity become takeover targets themselves. Therefore, firms seem to have an incentive to get their equity overvalued, so that they can make acquisitions with stocks. Servaes (1991) confirms this statement by finding that total returns are 10% larger in cash takeovers than in pure securities takeovers. Another determinant found by Servaes (1991) is deal atmosphere. He showed that the losses to acquiring firms are in general 4% larger in hostile takeovers than in friendly takeovers, while the target firms' gain are 10% more in a hostile takeover. This difference can be explained by the higher premium paid in hostile takeovers or because of the decrease in firm value caused by the takeover defenses.

Moeller and Schlingemann (2004) studied the influences of M&A types. In comparison with domestic M&As, they found that acquirers experience lower stock and operating performances in case of cross-border M&As. Therefore they conclude that a negative relation between value and global diversification exists. Another study by Moeller, Schlingemann and Stulz (2004) showed the size effect of the acquirer on the gains from acquisitions. In this study they found significant results which show higher announcement returns for large acquiring firms than for small acquiring firms.

Regarding business relatedness some contrary results are found. Sing and Montgomery (1987) found that acquisitions which are related in product, market or technological terms create substantially higher gains than unrelated acquisitions. This statement indicates that related target firms benefit more from acquisitions than unrelated target firms. However, Chatterjee (1986) found opposite results and shows that unrelated targets outperform the related targets.

Summarizing: When looking at the determinants of abnormal returns, various determinants and theories are discussed in the literature and their respective effects differ considerably as well.

3. Data and Methodology

This section presents the research design used to find the effects of M&As on firm performance. At this juncture the process of data collection will be discussed first, followed by the methodologies used regarding approach one (firm performance measured by stock prices) and approach two (firm performance measured by profits). The summary statistics per approach will be given after the description of the concerning methodology.

3.1. Data

To determine the effects of M&As on firm performance, this study uses data from three main sources. Data concerning M&As is taken from the Securities Data Company (SDC) Database, which covers transactions all over the world. Firms' financial data has been taken from the Compustat database and Datastream is used to obtain data regarding stock prices and country's market index return.

This study uses two different approaches to answer the research question. For both approaches the sample meets the following requirements. Only domestic transactions are taken into account (which constitute 82% of all public deals worldwide (Alexandridis et al. 2010)). The dataset contains data from the 2002-2013 period, the acquirer and target firm must be listed, the acquiring firm controls less than 50% of the shares of the target at announcement date and obtains 100% of the target shares after the transaction, the deal value is equal to or greater than 1 million dollar. Firms with multiple M&As and/or spin-/split-offs during the sample period and/or firms whose data is not completely available are excluded.

Countries are separated into six groups (which are the same for both approaches), based on the study of Gugler et al. (2002). These groups are: 1. United States of America, 2. Australia/New Zealand/Canada, 3. Japan, 4. United Kingdom, 5. Continental Europe, 6. Rest of the world. Continental Europe includes Austria, Belgium, Germany, Denmark, Spain, Finland, France, Greece, Ireland, Luxembourg, The Netherlands, Norway, Sweden, Portugal, Switzerland and Iceland. The Rest of the world sample includes 36 other countries around the world.

The original SDC dataset, that meets the sample requirements, consists of 3,568 completed M&As across the world. Almost half of these mergers (1,733) took place in the United States, 917 mergers took place in Australia/New Zealand/Canada, 291 mergers in Japan, 201 mergers in the United Kingdom, 117 in Continental Europe and 309 in the Rest of the world. Roughly 96% of the dataset contains friendly takeovers, 40% of all mergers are all-equity financed and the average deal value is 1,074 million dollar. Due to missing data for relevant variables, the samples used for the analysis are smaller than the number of observations in the original SDC database. In order to ensure that the samples are as large as possible, different datasets are composed for both approaches (starting both from the same original SDC database). The event study in approach 1 contains 3,319 observations and the analysis in approach 2 contains 2,287 observations.

3.2. Methodology approach 1: Firm performance measured by stock prices

3.2.1. Methodology Event study

The first approach uses an event study to determine the behaviour of stock prices around the M&A announcement dates. Event study methodology is a widely used method to determine the effects of M&As on stock price behavior (Brown and Warner, 1980). The effects will be measured by cumulative average abnormal returns (CAARs) during the event window. A positive CAAR suggests that the M&As have a positive effect on firm performance and a negative CAAR implies a negative effect on the firms.

Conducting an event study can be divided into three steps. In the first step the announcement date is determined using the information from the SDC database.[1] In the second step it is important to select a good benchmark model to determine abnormal stock returns. Several methods are available for selecting a good benchmark. In the mean-adjusted method, the benchmark is the average return over some period, where the benchmark period is slightly arbitrary. In the market adjusted returns method, market wide stock price movements are taken into account. Here the normal returns are calculated as the return on a market index. Using an equally weighted or a value weighted index makes no differences in this method (Brown and Warner, 1980). Finally, in the third step the abnormal returns are calculated and analyzed.

In this study the abnormal stock returns are determined for both, the bidder and the target firm. This is possible as one of the main initial selection criteria was that both the target and bidder were required to be listed. Standard event study methodology (Brown and Warner, 1980) is used and the event window consists of three days (following Moeller et al. 2005). A small event window is chosen because “the most statistically reliable evidence on whether mergers create value for shareholders comes from traditional short-window event studies” (Andrade et al. 2001). This is because a small event window reduces the chance that other effects influence the stock price. Abnormal returns are calculated by the following formula (Jong and Goeij, 2011):

$$AR_{it} = R_{it} - NR_{it}$$

Where R_{it} presents the stock return during the event window and where NR_{it} presents the stock return of the benchmark (normal return). To take into account the market wide stock price movements, this study uses market adjusted returns to estimate normal returns.

[1] Using the actual date would not make sense because at that moment the change in value due to the M&A should already be reflected in the stock price.

$$NR_{it} = R_{mt}$$

R_{mt} is found by the country's Datastream value-weighted market index return (following Alexandridis et al. 2010) and the estimation window consists of a [-205,-6] day interval (following Moeller et al. 2005). Within this window all daily market returns are calculated and the average is taken as normal return. Subsequently the cumulative abnormal return (CAR) and the CAAR are calculated (following Jong and Goeij, 2011).[2]

To be able to show the differences before and during the crisis, the CAARs for the sample per year and country in the 2002-2013 period are calculated and a t-test is used to find out whether the CAARs differ significantly from zero (following Jong and Goeij, 2011). This statistical test is necessary to determine whether the M&A event has a statistical influence on the stock price. The null hypothesis is that the expected average cumulative price change over the event window is zero. The alternative hypothesis expects the CAAR to be different from zero. The test statistic approximately follows a standard normal distribution for large N (>30) if the CARs are mutually uncorrelated.[3]

In case the sample is smaller than 30 observations, the assumption of normality is not valid. Therefore, this study uses the Wilcoxon signed-rank test to test the significance for samples with less than 30 observations. The Wilcoxon signed-rank test is a non-parametric statistical

[2] $CAR_i = \sum_{t=t_1}^{t_2} AR_{it}$
$CAAR = \frac{1}{N}\sum_{i=1}^{N} CAR_i$

[3] $H0: E(CAAR_{it}) = 0$
$H1: E(CAAR_{it}) \neq 0$

The t-test statistically formula is: $TS_2 = \sqrt{N}\frac{CAAR}{s} \approx N(0,1)$
Where N is the number of firms in the sample and s is the standard deviation:
$s = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(CAR_i - CAAR)^2}$

hypothesis test, which is used to test whether the sample median differs significantly from the hypothesized value. The test is used in several studies like the studies of DeFusco, Johnson and Zorn (1990) and Chang (1998). However, the Wilcoxon signed-rank test makes the assumptions that the variable is ordinal and has a symmetric distribution. When the result of the test is significant, the median of the CAAR is statistical different from zero and has an impact on the return of the acquiring or acquired firm.

3.2.2. Methodology Cross-sectional analysis

A cross-sectional analysis is used to explain the acquirer's and target's CARs. This regression is based on the cross-sectional analysis of CAR's widely used in event studies such as Braggion et al. (2012) and contains only the non-US firms from the event study. American companies are not taken into account because most studies were already focused on this country. This means only country groups 2, 3, 4, 5 and 6 are taken into account.[4] The regression equation used is the following:

$$CAR_{it} = \beta_0 + \beta_1 Payment\ in\ shares_{it} + \beta_2 ROE\ Acq_{it-1} + \beta_3 ROE\ Targ_{it-1} +$$
$$\beta_4 Targ\ Distress_{it} + Year\ Effects_t + Industry\ Effects_i + Country\ Effects_i + \varepsilon_{it}$$

Where CAR_{it} is the cumulative abnormal return of firm i in year t for the 2003-2013 window. The independent variables are determined as follows: Payment in shares equals 1 if the deal was financed only by issuing shares to target shareholders and 0 otherwise. Return on equity (ROE) is calculated as the previous year's profits (EBIT) divided by shareholder's equity. Target in Distress equals 1 if the target firm was in financial distress and 0 otherwise. To

[4] Country group 1 is United States, 2 is Australia/New Zealand/Canada, 3 is Japan, 4 is United Kingdom, 5 is Continental Europe and 6 is the Rest of the world.

determine whether a target firm was in financial distress, the Z-score is calculated (based on the study of Altman, 1968):

$$\text{Z-score} = A3.3 + B0.99 + C0.6 + D1.2 + E1.4$$

Where A is EBIT / Total assets, B is Sales / Total assets, C is Market value equity / Book value of total debt, D is Working capital / Total assets and E is Retained Earnings / Total Assets. A target firm with a Z-score below 1.81 is rated as a firm in financial distress (following Altman, 1968). Year fixed effects and Industry fixed effects are included in all regressions, Country fixed effects are included in the total sample regressions. The selection of industries is based on the divisions of the Standard Industrial Classification (SIC) codes. This is a system for classifying industries by a four-digit code. The number of divisions is reduced to three industries. Industry 1: Agriculture/Forestry/Fishing, Mining, Construction, Manufacturing and Transportation/Communication/Electric/Gas/ Sanitary Services. Industry 2: Wholesale Trade and Retail Trade. Industry 3: Finance/Insurance/Real Estate and Services. Finally, ε_{it} is an error term reflecting other factors that influence CAR_{it}.

3.2.3. Summary statistics approach 1

To find all required data for this regression, the Datastream database is combined with the Compustat database. Table 1 summarizes the descriptive statistics of the data used for the regression model. Due to missing variables in the Compustat database, the sample reduced from 3,319 observations in the original event study to 687 observations in the cross-sectional analysis. The average deal value amounts 836 million dollars. Looking at the deal values per country group, a remarkable high deal value of 2,359 million dollars is found in Continental Europe. This number can be explained by some high deal values realized during the sixth

Table 1. Summary statistics of the cross-sectional regression.

	Obs.	Mean	Std. Dev.	Min.	Max.
Total Sample					
Average deal value (Mn $)	687	836.20	3,211.93	1.68	37439.10
Payment method	687	0.61	0.49	0.00	1.00
ROE, acquirer	687	0.10	0.20	-1.12	1.36
ROE, target	687	0.03	0.26	-3.51	1.12
Target in distress	687	0.18	0.38	0.00	1.00
Australia/New Zealand/Canada					
Average deal value (Mn $)	201	352.90	1,304.16	1.96	15287.80
Payment method	201	0.43	0.50	0.00	1.00
ROE, acquirer	201	0.06	0.23	-1.03	0.87
ROE, target	201	0.01	0.42	-3.51	1.12
Target in distress	201	0.23	0.42	0.00	1.00
Japan					
Average deal value (Mn $)	189	629.29	1,623.23	1.68	9432.37
Payment method	189	0.87	0.33	0.00	1.00
ROE, acquirer	189	0.10	0.14	-0.17	1.36
ROE, target	189	0.05	0.15	-0.93	1.05
Target in distress	189	0.14	0.35	0.00	1.00
United Kingdom					
Average deal value (Mn $)	65	900.80	3,400.67	2.81	25439.40
Payment method	65	0.35	0.48	0.00	1.00
ROE, acquirer	65	0.07	0.19	-0.52	0.64
ROE, target	65	-0.02	0.09	-0.40	0.24
Target in distress	65	0.06	0.24	0.00	1.00
Continental Europe					
Average deal value (Mn $)	88	2,359.42	6,867.77	3.43	37439.10
Payment method	88	0.59	0.49	0.00	1.00
ROE, acquirer	88	0.14	0.23	-1.12	0.81
ROE, target	88	0.01	0.17	-0.60	0.72
Target in distress	88	0.25	0.44	0.00	1.00
Rest of world					
Average deal value (Mn $)	144	822.35	2,821.60	1.81	31756.70
Payment method	144	0.64	0.48	0.00	1.00
ROE, acquirer	144	0.16	0.20	-0.18	0.99
ROE, target	144	0.06	0.19	-0.88	0.73
Target in distress	144	0.17	0.38	0.00	1.00

Continental Europe includes Austria, Belgium, Germany, Denmark, Spain, Finland, France, Greece, Ireland, Luxembourg, The Netherlands, Norway, Sweden, Portugal, Switzerland and Iceland. The Rest of the world sample includes 36 other countries around the world. Deal value includes the amount paid by the acquirer, excluding fees and expenses. This dollar value includes the amount paid for all common stock, common stock equivalents, preferred stock, debt, options, assets, warrants, and stake purchases made within six months of the announcement date of the transaction. Liabilities assumed are included in the value if they are publicly disclosed. If a portion of the compensation paid by the acquirer is common stock, the stock is valued using the closing price on the last full trading day prior to the announcement of the terms of the stock swap. Payment in shares equals 1 if the deal was financed only by issuing shares to target shareholders and 0 otherwise. Return on Equity (ROE) equals the previous year's profits divided by shareholder's equity. Target in Distress equals 1 if the target firm was in financial distress and 0 otherwise (determined by calculating the firms' Z-score).

merger wave, which have a great influence on the average. As could be expected, the deal values during the financial crisis are much lower. Also the United Kingdom presents higher deal values than the other countries (901 million dollars). Australia/New Zealand/Canada, Japan and the Rest of the world realize lower deal values than the total sample average of 836 million dollars (353, 629 and 822 million dollars respectively). Roughly 61% of all M&As were entirely paid with shares. Japan is by far the country with the most all-equity financed M&As (87%), while the United Kingdom is the country with the least all-equity financed M&As (35%). This suggests that an overvalued market is more common in Japan than in the United Kingdom, because overvalued markets are an incentive to execute all-equity financed mergers (Schleifer and Vishny, 2003). Profitability, measured by ROE, is higher for acquiring firms (10%) than for target firms (3%). Specified per country group, the same trend is visible. In every country group the acquiring firm has a higher ROE than the target firm. The United Kingdom is the only country where the target firms realize on average a negative ROE (-2%). In only 18% of the M&As is the target firm in financial distress, which supports the conclusion of Ang and Mauck (2011) who found that acquiring firms do not benefit from acquiring distressed companies. The highest percentage of distressed firms is found in Continental Europe, where 25% of the target firms are in financial distress.

3.3. Methodology approach 2: Firm performance measured by profits

3.3.1. Methodology approach 2

The second approach follows closely the method developed in Gugler et al. (2002) and measures firm performance after an M&A by profits (EBIT). This means that not the most common method in literature (measuring profitability by computing weighted average profit rates of the merging firms before and after the merger (Mueller, 1997)) is used. Advantage of this deviating method is the possibility to find new insights regarding M&A effects with respect to existing studies and the possibility to re-examine the results of Gugler et al. (2002) in a more recent time period.

To determine whether a merger increased profits it is necessary to predict the profits of the two merging firms in the absence of the merger. Even if the merged firms realize a 25% increase in profits, it might be a failure because the merged firms' profits would have increased by 30% if they had not merged. To predict the performance of the merging firms in the absence of the merger a benchmark group of firms is chosen by industry. It is assumed that the merging firms' profits would have changed in the same way as the median firms' profits. The benchmark firms are all listed firms and are selected per country and SIC code on a two-digit level of aggregation to avoid problems with too small benchmark samples. Thereafter, equation 5 in Gugler et al. (2002) is used to calculate the projected profits of the combined company:

$$II_{Ct+n} = II_{Gt-1} + \frac{K_{IGt+n}}{K_{IGt-1}} K_{Gt-1} \Delta_{IGt-1,t+n} + II_{Dt} + \frac{K_{IDt+n}}{K_{IDt}} K_{Dt} \Delta_{IDt,t+n}$$

Where $\Delta_{IGt-1,t+n}$ is the projected change in the returns on the acquirer's asset from year *t-1* to t+n using the changes observed for the median company in its industry:

$$\Delta_{IGt-1,t+n} = \frac{II_{IGt+n}}{K_{IGt+n}} - \frac{II_{IGt-1}}{K_{IGt-1}}$$

Because profits can take positive, negative and zero values, changes in the ratios of profits to total assets are used to predict changes in the profits of the merging firms. In these formulas *t* is the year of the M&A, *n* is the number of years after the M&A and the variables are defined as: II_{ct+n}, predicted profits of the combined company in year *t+n*; II_{Gt+n}, profits of the acquiring firm in year t+n; K_{IGt+n}, assets of the median firm in the industry of the acquiring firm in year t+n; K_{Gt+n}, assets of the acquiring firm in year t+n; II_{Dt}, profits of the acquired firm in year t; K_{IDt+n}, assets of the median firm in the industry of the acquired firm in year t+n; K_{Dt}, assets of the acquired company in year *t*; $\Delta_{IDt,t+n}$, projected change in the return on assets of the acquired firm from year *t* to t+n (calculated analogously as $\Delta_{IGt-1,t+n}$); II_{IGt+n}, profits of the median firm in the industry of the acquiring firm in year *t.* According to the formula, the profits of the combined firms in year t+n are calculated as the profits of the acquiring firm in year *t-1*, plus the predicted growth in profits from years *t-1* to t+n, plus the profits of the acquired firm in year *t*, plus the predicted growth in profits from years *t* to t+n.

To determine the effect of M&As on profits, the projected performance is compared to the realized performance of the combined company. When the realized performance is higher than the projected performance, the M&A effect is positive and in case of a lower realized performance, the M&A effect is negative. To arrive at comparable real values all values regarding approach 2 are converted to USD and deflated by the US-Consumer Price Index with base year 2008. The level of significance is denoted by p-values and in cases of small sample sizes (N<30) by the Wilcoxon signed-rank test.

3.3.2. Summary statistics approach 2

Table 2 presents the summary statistics of the data used to measure firm performance by profits. Due to missing data and the elimination of outliers (the left and right 1% tail of the distribution are dropped) the number of deals from the original SDC database declined to 626 deals with an average deal value of 1,342 million dollars. Because of the change in sample, the average deal values from approach 2 differ from the values in approach 1. Per country group specified, Continental Europe has the highest deal value of 2,245 million dollars, which corresponds with the ranking found in the event study. The other countries differ in ranking, where the United Kingdom has the lowest deal value (253 million dollars). Japan, Australia/New Zealand/Canada, The rest of the world and the United States have an average deal value of 476, 593, 742 and 2,114 million dollars respectively.

The number of observations increases for the realized and projected profits, due to the measurement per year up to 5 years after the M&A. Based upon the realized and projected profits, it can be concluded that M&As in most cases result in higher profits. Over the total sample the realized profits are on average 870 million dollars, while the average projected profits in the absence of the merger amounts 795 million dollars. Japan is the only country that shows a negative influence of M&As on profitability. In this country the realized profits are 427 million dollars against projected profits in the absence of the merger of 435 million dollars. Firms in Continental Europe benefit the most from M&As. Their realized profits are on average 1,230 million dollars, while their projected profits are 923 million dollars. Most of the outcomes found in table 2 correspond with the study of Gugler et al. (2002), where also negative M&A effects for Japan and positive M&A effects for the United States, Europe, United Kingdom and the Rest of the world were found.

Table 2. Summary statistics approach 2.

In Mn $	Obs.	Mean	Std. Dev.	Min.	Max.
Total Sample					
Deal value	626	1,342.10	4,870.62	1.30	67,285.70
Realized profits	2,287	870.36	2,670.70	-11,982.00	35,872.30
Projected profits	2,287	794.72	2,458.45	-5,991.13	30,206.79
United States					
Deal value	254	2,114.40	6,255.15	2.77	67,285.70
Realized profits	920	1,181.31	3,266.96	-11,982.00	35,872.30
Projected profits	920	1,105.52	2,934.93	-5,991.13	30,206.79
Australia/New Zealand/Canada					
Deal value	96	592.80	1,973.97	1.76	17,932.98
Realized profits	351	651.29	3,017.77	-4,266.11	24,790.40
Projected profits	351	634.43	3,045.62	-2,888.25	27,145.23
Japan					
Deal value	133	475.73	1,332.66	1.68	9,148.93
Realized profits	530	427.28	1,464.97	-1,503.37	16,827.16
Projected profits	530	435.14	1,322.46	-315.19	11,595.36
United Kingdom					
Deal value	15	252.93	374.53	2.83	1,035.06
Realized profits	39	1,665.25	3,822.32	-32.30	12,613.25
Projected profits	39	1,431.71	2,763.74	-83.86	8,804.22
Continental Europe					
Deal value	51	2,245.33	8,529.08	1.30	60,856.45
Realized profits	193	1,229.57	2,163.61	-628.95	12,293.89
Projected profits	193	922.81	2,168.01	-2,183.81	13,842.03
Rest of world					
Deal value	77	741.51	1,251.69	3.12	8,796.79
Realized profits	254	576.31	1,316.64	-2,040.14	13,311.31
Projected profits	254	445.69	1,124.08	-1,464.45	8,332.31

Continental Europe includes Austria, Belgium, Germany, Denmark, Spain, Finland, France, Greece, Ireland, Luxembourg, The Netherlands, Norway, Sweden, Portugal, Switzerland and Iceland. The Rest of the world sample includes 36 other countries around the world. Deal value includes the amount paid by the acquirer, excluding fees and expenses. This dollar value includes the amount paid for all common stock, common stock equivalents, preferred stock, debt, options, assets, warrants, and stake purchases made within six months of the announcement date of the transaction. Liabilities assumed are included in the value if they are publicly disclosed. If a portion of the compensation paid by the acquirer is common stock, the stock is valued using the closing price on the last full trading day prior to the announcement of the terms of the stock swap. Realized profits are the realized EBITs of the combined firms. Projected profits are the expected profits of the merging firms in the absence of the merger.

4. Results

In this section the main results regarding effects of M&As on firm performance will be discussed per approach. First the results of the event study will be presented, followed by the results of the profit analysis.

4.1. Results approach 1: Firm performance measured by stock prices

4.1.1. Results Event study

Table 3 presents the results of the event study for the whole sample period. Due to missing Sedol and Ticker codes in the Compustat database relative to the SDC Database, it was not possible to perfectly match those two databases, which caused a difference in the number of target and acquiring firms. Regarding the total sample results, the target firms realize a positive CAAR of 16.43%, while the acquiring firms only realize a small positive return of 0.73%. Looking at the effects per country group, this trend is confirmed. The target firms realize in every country group a higher CAAR than the acquiring firms. All results, except the results of the acquiring firms in the U.K. and Continental Europe, are statistically significant. This finding, where target shareholders perform better than acquirer shareholders, confirms the conclusion from the literature review where the same trend was found in several studies like Andrade et al. (2001) and Franks and Harris (1989). The effects for target firms are largely positive, while the effects for the acquiring firms are just above zero. Though the result for the acquiring firms in the U.K. is not statistically significant, the effect for acquiring firms in this country seems to be even slightly negative (-0.17%).

Table 3. Firm performances in the 2002-2013 period, measured by stock price movements.

	Target	Acquirer		Target	Acquirer
Total Sample					
Number of observations	1,753	1,566			
CAAR(-1,+1)	0.1643	0.0073			
% Positive	79.98%	51.92%			
T-value	30.74***	4.18***			
United States			United Kingdom		
Number of observations	927	712	Number of observations	116	111
CAAR(-1,+1)	0.2119	0.0043	CAAR(-1,+1)	0.1062	-0.0017
% Positive	84.68%	53.09%	% Positive	77.59%	45.95%
T-value	26.27***	1.66*	T-value	6.43***	-0.38
Australia/New Zealand/Canada			Continental Europe		
Number of observations	372	401	Number of observations	64	69
CAAR(-1,+1)	0.1574	0.0112	CAAR(-1,+1)	0.0910	0.0039
% Positive	78.76%	49.88%	% Positive	70.31%	49.28%
T-value	14.07***	2.79***	T-value	4.40***	0.52
Japan			Rest of world		
Number of observations	169	153	Number of observations	105	120
CAAR(-1,+1)	0.0605	0.0126	CAAR(-1,+1)	0.0446	0.0158
% Positive	67.46%	54.90%	% Positive	71.43%	55.00%
T-value	7.13***	2.76***	T-value	4.60***	2.64***

CAAR(-1,+1) denotes the 3-day cumulative average abnormal return (in percent). % positive is the percentage of positive CARs during the event window. T-value is the probability that the observed CAAR is significantly different from zero.

* Statistical significance at the 10% level.

** Statistical significance at the 5% level.

*** Statistical significance at the 1% level.

The percentages positive returns show that over the total sample 80% of the CARs are positive for the target firms and 52% for the acquiring firms. Per country specified, the percentage positive returns for the target firms is the lowest in Japan (67%). The percentages regarding acquiring firms show in three country groups, Australia/New Zealand/Canada, U.K. and Continental Europe, a number lower than 50% (49.9%, 46% and 49% respectively). Hence, the percentages positive returns confirm that target firms perform better than acquiring firms. They realize in all country groups more positive returns than negative returns, while acquiring firms realize more negative than positive returns in three country groups.

Looking at the effects concerning target firms, the highest return is found for the U.S. (21.19%), followed by Australia/New Zealand/Canada with 15.74%, the U.K. with 10.62% and Continental Europe, Japan and the Rest of the world with 9.1%, 6.05% and 4.46% respectively. These results are endorsed by the study of Park (2004) who found significant greater abnormal returns for U.S. firms than non-U.S. firms. Regarding the acquiring firms, a different ranking is found. Rest of the world has the highest return (1.58%) followed by Japan (1.26%), Australia/New Zealand/Canada (1.12%), U.S. (0.43%), Continental Europe (0.39%) and U.K. (-0.17%), although the numbers of Continental Europe and U.K. are statistically insignificant. The results of Continental Europe correspond almost identically with the results of Campa and Hernando (2004), who found cumulative abnormal returns of 9% for target shareholders and no cumulative abnormal returns for the shareholders of acquiring firms. Therefore, the trend found in literature regarding abnormal returns is confirmed in table 1, where the CAAR is largely positive for target firms and much lower, around zero, for acquiring firms. The U.S. shows the highest return for target firms, while the Rest of the world country group shows in this category the lowest return. On the other hand, the Rest of the world country group shows the highest return regarding acquiring firms, where the U.S. owns the fourth place in ranking.

To be able to show the differences before and during the crisis, the CAARs for the sample per year, per period and per country in the 2002-2013 period are given in tables 4 and 5. Both, T-values and Wilcoxon Z-values are reported to measure the level of significance. T-values are used for samples larger than 30 observations and Wilcoxon Z-values are used for samples with less than 30 observations. The pre-crisis period contains the years 2002-2007 and the crisis period contains the years 2008-2013. Tables 4 and 5 present the results of the target and acquiring firms, where in both tables a decrease in sample size in the crisis period with respect to the pre-crisis period has been occurred. This decline in sample size was also found by Gaughan (2009) and is probably caused by the problem of credit accessibility and the decline in sales and profits, which make it unattractive to execute M&As during the crisis.

Table 4 presents the results for the target firms. Looking at the results of the total sample it strikes that the outcomes of the crisis-period show significant higher CAARs than the outcomes in the pre-crisis period (19.71% against 14.05%). Also the percentage positive returns is higher in this period (82% against 79%). Specifying these results per year, the lowest CAAR (13.68%) in the crisis period can be found in 2009. But this value is still higher than the values in 2004 (10.96%), 2005 (12.8%) and 2006 (12.73%). Results per country show the same trend and are statistically significant at the 1% level (only the pre-crisis period of the Rest of the world sample is significant at the 5% level). The percentages positive returns show all high values (higher than 67%) in both periods and for all country groups, but do not always show the same trend. The percentages of the crisis period are sometimes surpassed by the percentages of the pre-crisis period (referring to Japan and the U.K.).

Table 4. Yearly firm performances of target firms, measured by stock price movements.

Year of announcement	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	Pre-crisis period	Crisis period
Total Sample														
Number of observations	140	156	178	176	176	190	118	131	133	124	116	115	1,016	737
CAAR(-1,+1)	0.1917	0.1461	0.1096	0.1280	0.1273	0.1508	0.1571	0.1368	0.2326	0.1917	0.2373	0.2313	0.1405	0.1971
% Positive	78.57%	82.05%	75.28%	77.84%	78.41%	79.47%	77.97%	77.86%	83.46%	81.45%	86.21%	85.22%	78.50%	81.95%
T-value	7.85***	9.18***	9.14***	9.81***	9.81***	10.87***	7.32***	7.65***	9.31***	9.44***	9.88***	9.23***	22.49***	21.38***
Wilcoxon Z	8.05***	8.83***	8.53***	8.70***	9.15***	9.96***	7.27***	7.39***	8.60***	8.02***	8.27***	8.23***	21.75***	19.54***
United States														
Number of observations	80	102	102	100	91	105	54	55	68	47	55	68	580	347
CAAR(-1,+1)	0.2642	0.1776	0.1471	0.1734	0.1646	0.1879	0.2215	0.2242	0.3238	0.2319	0.2985	0.2425	0.1833	0.2597
% Positive	81.25%	85.29%	78.43%	84.00%	81.32%	85.71%	83.33%	87.27%	91.18%	87.23%	92.73%	85.29%	82.76%	87.90%
T-value	6.98***	8.22***	8.06***	9.58***	7.96***	9.54***	6.10***	6.45***	8.27***	7.27***	8.58***	7.41***	19.73***	17.79***
Wilcoxon Z	6.70***	7.63***	7.02***	7.65***	7.09***	8.08***	5.57***	5.77***	6.75***	5.49***	6.21***	6.41***	18.05***	14.78***
Australia/New Zealand/Canada														
Number of observations	14	14	34	28	41	47	40	31	32	38	26	27	178	194
CAAR(-1,+1)	0.1710	0.1755	0.0621	0.0942	0.1098	0.1183	0.1226	0.1182	0.2097	0.2528	0.2271	0.2994	0.1105	0.2004
% Positive	78.57%	85.71%	79.41%	67.86%	73.17%	72.34%	82.50%	74.19%	87.50%	81.58%	76.92%	92.59%	74.72%	82.47%
T-value	2.46**	3.73***	4.63***	2.54**	5.31***	4.43***	5.16***	4.19***	4.82***	5.91***	3.65***	5.46***	8.79***	11.41***
Wilcoxon Z	2.35**	2.98***	3.96***	2.23**	4.43***	4.21***	4.52***	3.33***	4.66***	4.71***	3.44***	4.47***	8.36***	10.32***
Japan														
Number of observations	24	15	19	20	19	8	8	10	8	15	17	6	105	64
CAAR(-1,+1)	0.0769	0.0682	0.0507	0.0285	0.0429	0.0162	0.0550	0.0326	0.1164	0.0638	0.1274	0.0107	0.0509	0.0763
% Positive	70.83%	80.00%	63.16%	55.00%	73.68%	62.50%	62.50%	70.00%	87.50%	60.00%	70.59%	50.00%	67.62%	67.19%
T-value	3.13***	2.74***	2.16**	1.31	2.13**	1.63	1.54	1.33	2.45**	1.73*	3.91***	0.23	5.27***	4.84***
Wilcoxon Z	2.80***	2.33**	2.09**	1.05	2.05**	1.12	1.26	1.27	1.96**	1.19	2.86***	0.11	4.96***	3.77***
United Kingdom														
Number of observations	12	12	9	18	13	14	6	8	8	6	4	6	78	38
CAAR(-1,+1)	0.0623	0.0872	0.0809	0.0770	0.1466	0.1419	0.1956	0.0416	0.1083	0.0625	0.1889	0.1688	0.1000	0.1188
% Positive	83.33%	83.33%	44.44%	83.33%	100.00%	78.57%	50.00%	62.50%	50.00%	83.33%	100.00%	100.00%	80.77%	71.05%
T-value	2.50**	2.35**	1.30	3.24***	3.00***	2.72***	1.08	0.99	1.77*	1.35	2.02**	1.55	6.00***	3.19***
Wilcoxon Z	2.12**	2.51**	0.77	3.11***	3.18***	2.73***	0.31	0.84	0.84	1.36	1.83*	2.20**	6.00***	3.24***
Continental Europe														
Number of observations	8	6	8	4	4	8	4	6	2	6	4	4	38	26
CAAR(-1,+1)	0.0729	-0.0191	0.0838	0.0618	0.0323	0.1178	0.0283	0.0313	0.0365	0.0894	0.4203	0.1938	0.0647	0.1295
% Positive	75.00%	50.00%	87.50%	50.00%	50.00%	75.00%	75.00%	50.00%	100.00%	66.67%	100.00%	75.00%	68.42%	73.08%
T-value	1.53	-0.84	4.67***	0.86	0.60	2.51**	0.64	0.97	1.72*	1.11	2.80***	1.49	3.59***	3.01***
Wilcoxon Z	1.40	-0.31	2.38**	0.37	0.00	1.82*	0.73	0.73	1.34	0.73	1.83*	1.46	3.04***	2.78***
Rest of world														
Number of observations	2	7	6	6	8	8	6	21	15	12	10	4	37	68
CAAR(-1,+1)	0.0621	0.0375	0.0036	0.0589	0.0100	0.0371	-0.0080	0.0516	0.0223	0.1164	0.0603	0.0429	0.0308	0.0520
% Positive	50.00%	57.14%	66.67%	100.00%	62.50%	62.50%	50.00%	76.19%	53.33%	91.67%	90.00%	75.00%	67.57%	73.53%
T-value	0.53	0.85	0.09	2.35**	0.76	1.23	-0.26	3.21***	0.71	2.55**	2.67***	1.52	2.25**	4.02***
Wilcoxon Z	0.45	0.68	0.11	2.20**	0.58	0.26	-0.11	2.73***	0.23	2.90***	1.99**	1.46	2.15**	4.16***

CAAR(-1,+1) denotes the 3-day cumulative average abnormal return (in percent). % positive is the percentage of positive CARs during the event window. T-value and the nonparametric Wilcoxon Z-value report the probability that the observed CAAR is significantly different from zero. The Pre-crisis period includes the years 2002-2007, the Crisis period includes the years 2008-2013.

* Statistical significance at the 10% level.

** Statistical significance at the 5% level.

*** Statistical significance at the 1% level.

The country group ranking in the pre-crisis period is similar to the crisis period and corresponds with the ranking over the whole sample period in table 1. This means that the U.S. shows the highest CAARs for both the pre-crisis and crisis period (18.33% and 25.97% respectively), while the Rest of the world country group realizes the lowest CAARs (3.08% in the pre-crisis period and 5.2% in the crisis period). Looking at the results per year, it can be observed that the trend found in the pre-crisis and crisis period does not apply for every year. For example, the country group Australia/New Zealand/Canada shows in 2008 and 2009 relative small CAARs (12.26% and 11.82% respectively) in comparison with the years after. Consequently, some years in the pre-crisis period show higher CAARs (17.1% in 2002, 17.55% in 2003 and 11.83% in 2007). Also a couple of other country groups show some deviating years, but hereby should be noted that not all results per year are statistically significant. Therefore, from table 2 it can be concluded that M&A announcements in the crisis period provide higher CAARs for target firms than M&A announcements in the pre-crisis period. This trend can be observed for all country groups. Again the highest returns are found for the U.S., followed by Australia/New Zealand/Canada, U.K / Continental Europe, Japan and the Rest of the world.

The findings presented in table 5 show the effects of M&A announcements for acquiring firms. Although the results in significance levels are mixed, the same trend in CAARs can be found as in table 4. Looking at the total sample, the CAAR in the pre-crisis period is 0.27% and in the crisis-period 1.56%. Also the percentage positive returns is higher in the crisis period than in the pre-crisis period (54% and 51% respectively) indicating that the market more favourably values M&As during the crisis period. Specifying the results per country, all countries, except Continental Europe and the U.K., show the same trend. However, Continental Europe and the U.K. are the only country groups with statistically insignificant

Table 5. Yearly firm performances of acquiring firms, measured by stock price movements.

Year of announcement	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	Pre-crisis period	Crisis period
Total Sample														
Number of observations	191	168	163	161	159	162	107	121	90	90	74	80	1004	562
CAAR(-1,+1)	0.0050	0.0018	0.0005	-0.0006	0.0047	0.0042	-0.0099	0.0276	0.0125	0.0147	0.0295	0.0231	0.0027	0.0156
% Positive	53.93%	53.57%	52.15%	45.96%	52.20%	46.91%	46.73%	52.07%	54.44%	53.33%	59.46%	60.00%	50.90%	53.70%
T-value	1.22	0.49	0.11	-0.15	1.05	0.76	-1.61	2.64***	1.97**	1.53	2.94***	2.66***	1.50	4.26***
Wilcoxon Z	1.26	0.31	-0.09	-0.75	0.65	-0.84	-1.41	1.66*	1.74*	0.78	2.14**	2.23**	0.35	2.85
United States														
Number of observations	111	97	86	78	71	67	40	33	37	26	26	40	510	202
CAAR(-1,+1)	0.0044	-0.0019	-0.0049	-0.0055	0.0026	-0.0011	-0.0124	0.0252	0.0149	0.0200	0.0213	0.0374	-0.0008	0.0171
% Positive	58.56%	49.48%	51.16%	47.44%	53.52%	46.27%	40.00%	51.52%	64.86%	57.69%	57.69%	70.00%	51.57%	56.93%
T-value	0.80	-0.40	-0.71	-1.06	0.41	-0.19	-1.15	0.97	1.70*	0.77	1.50	3.04***	-0.36	2.54**
Wilcoxon Z	1.47	-0.44	-0.87	-0.95	0.25	-0.82	-1.18	0.26	1.92*	0.34	1.23	2.98***	-0.36	2.21**
Australia/New Zealand/Canada														
Number of observations	27	24	38	40	41	53	37	46	24	28	22	21	223	178
CAAR(-1,+1)	0.0106	0.0035	0.0126	-0.0040	0.0132	0.0117	-0.0132	0.0365	0.0093	0.0138	0.0324	0.0062	0.0083	0.0149
% Positive	40.74%	58.33%	55.26%	37.50%	53.66%	49.06%	48.65%	50.00%	50.00%	53.57%	63.64%	42.86%	48.88%	51.12%
T-value	0.82	0.34	1.43	-0.49	1.17	0.85	-1.32	2.07**	0.73	0.99	1.51	0.30	1.74*	2.18**
Wilcoxon Z	0.02	0.43	1.21	-1.30	0.85	-0.08	-1.18	1.33	0.51	0.64	1.06	-0.43	0.44	0.88
Japan														
Number of observations	26	17	18	18	17	10	5	9	7	12	10	4	106	47
CAAR(-1,+1)	0.0152	0.0027	0.0105	0.0038	0.0128	-0.0113	0.0252	0.0364	0.0264	0.0011	0.0415	0.0157	0.0076	0.0240
% Positive	50.00%	52.94%	66.67%	55.56%	64.71%	30.00%	60.00%	77.78%	42.86%	41.67%	60.00%	50.00%	54.72%	55.32%
T-value	1.33	0.23	1.51	0.50	0.93	-1.00	0.79	1.95*	1.28	0.06	1.14	0.54	1.68*	2.22**
Wilcoxon Z	0.95	0.12	1.33	0.63	0.88	-0.97	0.41	1.72*	0.68	-0.71	0.46	0.00	1.30	1.41
United Kingdom														
Number of observations	13	13	9	17	12	12	7	11	5	5	2	5	76	35
CAAR(-1,+1)	-0.0065	0.0265	-0.0279	0.0101	-0.0029	-0.0046	-0.0420	0.0019	-0.0266	0.0051	-0.0251	0.0307	0.0012	-0.0079
% Positive	61.54%	69.23%	22.22%	41.18%	33.33%	33.33%	42.86%	36.36%	40.00%	80.00%	0.00%	80.00%	44.74%	48.57%
T-value	-0.62	1.56	-1.90*	1.02	-0.40	-0.81	-1.54	0.17	-0.83	0.35	-7.07***	1.14	0.24	-0.85
Wilcoxon Z	-0.04	1.08	-1.72*	0.17	-0.39	-0.78	-0.85	-0.18	-0.67	0.67	-1.34	0.94	-0.56	-0.34
Continental Europe														
Number of observations	10	7	6	3	7	6	6	6	3	6	4	5	39	30
CAAR(-1,+1)	-0.0046	0.0088	0.0339	0.0116	-0.0264	0.0387	0.0016	-0.0321	0.0862	-0.0107	0.0307	-0.0338	0.0077	-0.0012
% Positive	50.00%	71.43%	66.67%	66.67%	42.86%	50.00%	50.00%	33.33%	66.67%	33.33%	50.00%	20.00%	56.41%	40.00%
T-value	0.29	0.63	1.31	0.77	-1.46	0.89	0.07	-2.00**	1.40	-0.48	0.83	-2.78***	0.79	-0.10
Wilcoxon Z	-0.05	0.85	1.15	0.54	-1.18	0.31	0.11	-1.57	1.07	-0.31	0.73	-1.75*	0.61	-0.57
Rest of world														
Number of observations	4	10	6	5	11	14	12	16	14	13	10	5	50	70
CAAR(-1,+1)	-0.0188	-0.0055	-0.0208	0.0455	0.0025	0.0049	0.0065	0.0419	0.0027	0.0341	0.0426	0.0354	0.0014	0.0262
% Positive	25.00%	50.00%	33.33%	60.00%	45.45%	64.29%	58.33%	62.50%	42.86%	53.85%	70.00%	80.00%	50.00%	58.57%
T-value	-1.05	-0.54	-1.23	1.64*	0.20	0.32	0.41	1.64*	0.18	1.54	1.77*	1.94*	0.20	2.93***
Wilcoxon Z	-1.10	-0.56	-1.15	1.21	0.09	0.35	0.39	1.60	-0.35	1.15	1.58	1.75*	0.02	2.50**

CAAR(-1,+1) denotes the 3-day cumulative average abnormal return (in percent). % positive is the percentage of positive CARs during the event window. T-value and the nonparametric Wilcoxon Z-value report the probability that the observed CAAR is significantly different from zero. The Pre-crisis period includes the years 2002-2007, the Crisis period includes the years 2008-2013.

* Statistical significance at the 10% level.

** Statistical significance at the 5% level.

*** Statistical significance at the 1% level.

results for both the pre-crisis and the crisis period. Regarding the results per year, the outcomes are in most cases statistically insignificant for all country groups. However, the outcomes suggest several negative values for acquiring firms. This finding would confirm the conclusion of Andrade et al. (2001) who also found negative abnormal returns for the acquiring firms around the announcement date.

Regarding the ranking of CAAR values, Australia/New Zealand/Canada is the country group with the highest value in the pre-crisis period (0.83%). In the crisis period the highest value is realized by the Rest of the world country group (2.62%), which corresponds with the findings from table 1. The U.S. realizes the lowest CAAR in the pre-crisis period (-0.08%) and the U.K. is responsible for the lowest value in the crisis period (-0.79%). Thus, although the mixed results in significance levels, the trend regarding acquiring firms correspond with the trend found regarding target firms. Therefore, also acquiring firms seem to realize higher CAARs in the crisis period than in the pre-crisis period. A reason to explain this trend could be the decline in the number of M&As. Most firms take fewer risks during a financial crisis and consider an M&A therefore better before actually undertaking one. This results in higher CAARs for both target and acquiring firms. Another reason could be that firms perform worse during the crisis and an M&A announcement is more positively valued in such a time as it can result in a turnaround for struggling firms. However, to make those conclusions, a study to the causes of higher CAARs during an economic crisis is required.

4.1.2. Results Cross-sectional regression

To explain the CARs a cross-sectional analysis is used for both target and acquiring firms. In order to do this, the CARs are regressed on various characteristics. The results for the target firms are presented in table 6. Due to the decline in sample size, the sample of target firms in country group 4 (U.K.) became too small and is therefore omitted in this cross-sectional analysis specified per country.

Table 6. Cross-sectional analysis of target's cumulative abnormal returns.

Country group	All	2	3	5	6
Constant	0.197***	0.100	0.140**	0.447**	0.075**
	(0.049)	(0.200)	(0.067)	(0.217)	(0.031)
Payment in Shares	-0.056**	0.008	0.034	-0.205*	0.011
	(0.026)	(0.060)	(0.056)	(0.123)	(0.026)
ROE, Acquirer	0.152*	0.140	0.233	-0.275	0.225*
	(0.079)	(0.135)	(0.215)	(0.475)	(0.120)
ROE, Target	-0.163***	-0.156***	-0.065	0.091	-0.084
	(0.035)	(0.056)	(0.083)	(0.208)	(0.058)
Target in Distress	-0.018	0.036	-0.036	-0.090	-0.004
	(0.027)	(0.059)	(0.037)	(0.075)	(0.026)
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes
Country fixed effects	Yes	No	No	No	No
Adjusted R^2	0.145	0.174	0.078	-0.126	0.272
Observations	192	68	60	24	33

Country group 2 is Australia/New Zealand/Canada, 3 is Japan, 5 is Continental Europe and 6 is the Rest of the world. The dependent variable is the target's CAR. Payment in shares equals 1 if the deal was financed only by issuing shares to target shareholders and 0 otherwise. Return on Equity (ROE) equals the previous year's profits divided by shareholder's equity. Target in Distress equals 1 if the target firm was in financial distress and 0 otherwise (determined by calculating the firm's Z-score). Standard errors appear in parentheses. Coefficients for the regression fixed effects are suppressed.[5]

* Statistical significance at the 10% level.

** Statistical significance at the 5% level.

*** Statistical significance at the 1% level.

The most significant results are found in the first regression, which includes data from country groups 2, 3, 4, 5, and 6. This regression shows that the higher the acquirer's ROE, the higher is the target's CAR (0.152). Regarding the target's ROE, the effect is reverse, the lower the

[5] The results are similar when the standard errors are clustered on industry.

target’s ROE, the higher is the target’s CAR (-0.163). These results are statistically significant at the 10% and 1% level respectively and reflect the ability of incoming management to turn around a poorly performing target firm. Determining the economic significances of these independent variables, it can be concluded that a one standard deviation increase in a Target’s and Acquirer’s ROE results in a lower and higher CAR of -6.1% and 2.4% respectively. In comparison with a mean CAR of 8.1% it is clear that the change of the Target’s ROE is much larger. The result regarding payment in shares shows that all-equity financed M&As have a negative influence on the target’s CAR (-0.056). The variable Target in distress is statistically insignificant, but seems to have a small negative influence on the target’s CAR (-0.018).

Looking at the results per country group, roughly the same effect, which means the lower the target’s ROE, the higher is the target’s CAR, can be found for Australia/New Zealand/Canada (-0.156). To illustrate the large magnitude of this variable, note that a one standard deviation increase in a Target’s ROE reduces the CAR from its mean of 14.5% with -8.4%. Japan and the Rest of the world show also a negative effect, but these two results are statistically insignificant. Results in significance levels are also mixed for the variable ROE acquirer, but it seems to be that Continental Europe deviates here from the other countries. The coefficient has a negative value (-0.275) in this country group, while other countries present positive values. The variable Payment in shares is only statistically significant in Continental Europe and shows here a negative influence of -0.205, which means that an all-equity financed M&A in Continental Europe results in a lower CAR for the target firm. The results regarding the variable Target in Distress are all statistically insignificant, but suggest negative influences in Japan, Continental Europe and the Rest of world. In Australia/New Zealand/Canada this variable seems to have a positive effect.

Table 7. Cross-sectional analysis of acquirer's cumulative abnormal returns.

Country group	All	2	3	4	5	6
Constant	0.074***	0.059*	0.138***	-0.038	0.035	0.070***
	(0.013)	(0.035)	(0.025)	-0.029	(0.039)	(0.024)
Payment in Shares	-0.002	0.001	-0.006	-0.007	-0.017	0.002
	(0.007)	(0.019)	(0.016)	(0.012)	(0.018)	(0.014)
ROE, Acquirer	-0.031**	-0.086**	-0.065	0.061*	-0.035	-0.002
	(0.016)	(0.041)	(0.041)	(0.032)	(0.035)	(0.034)
ROE, Target	0.017	0.033	-0.018	-0.017	0.127*	0.007
	(0.016)	(0.058)	(0.052)	(0.138)	(0.068)	(0.047)
Target in Distress	-0.010	-0.014	-0.036*	0.028	0.017	-0.012
	(0.010)	(0.025)	(0.019)	(0.044)	(0.024)	(0.023)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Country fixed effects	Yes	No	No	No	No	No
Adjusted R^2	0.035	0.004	0.162	0.043	0.125	0.026
Observations	495	133	129	58	64	111

Country group 2 is Australia/New Zealand/Canada, 3 is Japan, 4 is United Kingdom, 5 is Continental Europe and 6 is the Rest of the world. The dependent variable is the acquirer's CAR. Payment in shares equals 1 if the deal was financed only by issuing shares to target shareholders and 0 otherwise. Return on Equity (ROE) equals the previous year's profits divided by shareholder's equity. Target in Distress equals 1 if the target firm was in financial distress and 0 otherwise (determined by calculating the firm's Z-score). Standard errors appear in parentheses. Coefficients for the regression fixed effects are suppressed.[6]

* Statistical significance at the 10% level.

** Statistical significance at the 5% level.

*** Statistical significance at the 1% level.

Table 7 presents the results regarding the acquirer's CAR. The first column contains all non-US firms and shows opposite results in comparison with the ROE variables in table 6. The higher the acquirer's ROE, the lower the acquirer's CAR (-0.031). Looking at the economic importance of this variable, it can be concluded that a one standard deviation increase in the acquirer's ROE results in a lower CAR of -0.67%, while the mean CAR in this regression is 1.8%. For the target ROE an opposite effect is found, indicating that a higher ROE of the

[6] The results are similar when the standard errors are clustered on industry.

target firm leads to a higher CAR for the acquiring firm (0.017), although this result is statistically insignificant. Also the results of the variables Payment in shares and Target in distress are statistically insignificant, but seem to have both negative influences. This would confirm the findings from the literature review, where a negative relation between Payment in shares and total returns was found by Servaes (1991) and where it appeared that acquiring firms do not benefit from acquiring distressed companies (Ang and Mauck, 2011).

The regressions specified per country show small negative effects for all-equity financed M&As in Japan, U.K. and Continental Europe. For Australia/New Zealand/Canada and the Rest of the world small positive effects are presented, although none of these results are statistically significant. The variable ROE acquirer shows statistically significant results in Australia/New Zealand/Canada and the U.K., where the effect in Australia/New Zealand/Canada is negative (-0.086) and in the U.K. positive (0.061). The magnitudes of the effects, measured by a one standard deviation increase in the independent variable, are -2.1% and 1.2% respectively. The mean CAR for Australia/New Zealand/Canada is 3% and the mean CAR for the U.K. is 1.5%. This means that the magnitudes of the effects are quite large. ROE target is statistically significant in Continental Europe and shows a positive effect of 0.127. Target in distress is negative and statistically significant in Japan (-0.036). In other countries the results are mixed, where the effect seems to be positive for the U.K. and Continental Europe and negative for Australia/New Zealand/Canada and the Rest of the World.

4.2. Results approach 2: Firm performance measured by profits

Approach 2 includes a profit analysis to see whether M&As on average have increased profits or reduced them. This second approach uses effective M&A dates instead of announcement dates. Because the Gugler formula needs data from the year prior to the M&A, the first

M&As used for this analysis took place in 2003. The size of the sample declines while moving away from the M&A date because companies disappear from the dataset. This is partly due to missing variables in the Compustat database and partly due to the fact that 2013 is the last year of the database. This means that M&As having taken place in 2008 are the last M&As for which data is available until year *t+5*, M&As having taken place in 2009 are in the sample only up to year *t+4*, M&As of year 2010 up to year *t+3* and so on. The profitability numbers are the average differences between the actual profits of the combined firm and its projected profits in year *t+n*. This means that a negative number implies a decline in profits and a positive number an increase in profits.

Table 8. Firm performances in the 2003-2013 period, measured by profits.

Years after the M&A	Number of observations	Profits: Difference in Mn $	% positive	P-value	Wilcoxon Z	Years after the M&A	Number of observations	Profits: Difference in Mn $	% positive	P-value	Wilcoxon Z
Total sample											
t+1	626	73.16	58.95%	0.035**	3.953***						
t+2	537	93.37	57.36%	0.012**	3.875***						
t+3	444	78.18	56.76%	0.092*	2.240**						
t+4	367	102.28	56.13%	0.027**	2.519**						
t+5	313	15.30	55.91%	0.762	1.727*						
United States						United Kingdom					
t+1	254	75.99	61.02%	0.259	2.462**	t+1	15	18.17	60.00%	0.940	-0.114
t+2	213	121.35	58.22%	0.023**	3.034***	t+2	11	175.76	54.55%	0.682	0.445
t+3	182	47.12	52.75%	0.515	0.199	t+3	6	82.32	66.67%	0.943	0.524
t+4	144	57.41	52.78%	0.488	1.023	t+4	4	798.51	50.00%	0.435	0.000
t+5	127	60.93	54.33%	0.498	1.420	t+5	3	1071.33	66.67%	0.477	0.535
Australia/New Zealand/Canada						Continental Europe					
t+1	96	56.22	61.46%	0.127	2.850***	t+1	51	164.08	56.86%	0.359	1.256
t+2	81	12.49	56.79%	0.887	1.076	t+2	44	276.74	61.36%	0.168	1.214
t+3	68	73.15	64.71%	0.547	1.870*	t+3	39	463.89	71.79%	0.084*	2.205**
t+4	59	8.88	64.41%	0.881	1.177	t+4	34	531.31	64.71%	0.047**	1.821*
t+5	47	-127.41	55.32%	0.399	-0.413	t+5	25	100.23	52.00%	0.614	0.336
Japan						Rest of world					
t+1	133	15.34	52.63%	0.723	0.414	t+1	77	135.28	61.04%	0.103	1.719*
t+2	122	18.83	50.82%	0.752	0.221	t+2	66	104.12	65.15%	0.402	2.514**
t+3	107	-25.61	50.47%	0.603	0.308	t+3	42	126.53	61.90%	0.111	1.957*
t+4	88	-18.37	45.45%	0.567	-0.395	t+4	38	239.59	73.68%	0.112	3.212***
t+5	80	-51.85	52.50%	0.402	0.336	t+5	31	47.40	74.19%	0.669	2.234**

Difference in Mn $ is the difference between actual and projected profits. A positive number therefore implies that the M&As increased profits, a negative number implies that the M&As decreased profits. P-value and Wilcoxon Z-value report the probability that the observed differences are significantly different from zero (two-sided test). % positive is the percentage of positive differences between actual and projected values, t is the year of the M&A.

* Statistical significance at the 10% level.

** Statistical significance at the 5% level.

*** Statistical significance at the 1% level.

Table 8 presents the findings for the whole sample period. For the total sample the mean difference between actual and projected profits is positive in all five years after the M&A. These results, which are statistically significant for the first four post-merger years, confirm the findings of Healy et al. (1992) and Gugler et al. (2002), who also showed positive effects of M&As on the profitability of merged companies. Also the percentages positive returns are in all years after the M&A above 50%, which means that a majority of M&As lead to higher actual profits than those predicted.

The figures per country group show that the U.S. realizes higher than predicted profits in every post-merger year, although only one of the five differences is statistically significant at the 5% level. The same trend can also be observed for the U.K., Continental Europe and the Rest of the world. Actual profits are in those country groups greater than projected profits in all five post-merger years. However, the differences are only statistically significant in Continental Europe in the three and four years after the M&A and the results are based on a quite small sample for the United Kingdom. The results for Australia/New Zealand/Canada and Japan seem to be different than those already discussed. Although these results are also statistically insignificant, in Australia/New Zealand/Canada the difference in year *t+5* seems to be negative (-127.41) and in Japan the differences seem to be negative in three of the five post-merger years (-25.61 in year *t+3*, -18.37 in year *t+4* and -51.85 in year *t+5*). Regarding the percentages positive differences, all country groups show in every post-merger year a percentage of at least 50%. Only year *t+4* of Japan is an exception with more negative than positive differences (here is only 45% of the differences positive). The results found so far correspond largely with the results of Gugler et al. (2002), which means that the effects of M&As on profitability did not changed a lot over time. Also Gugler et al. (2002) found positive returns for the U.S., the U.K., Continental Europe and the Rest of the world. Regarding Australia/New Zealand/Canada and Japan they found also both positive and negative differences in the post-merger years.

Making a ranking is difficult due to the statistically insignificant outcomes, but it confirms the findings of Mueller (1997). He summarized the results of 20 studies and concluded that most studies find statistically insignificant results when investigating the profitability of M&As per country. However, the results suggest that in the first three years after the M&A, the highest positive differences can be found in Continental Europe. This means that M&As in Continental Europe seem to be the most profitable. Also the U.K., although its small sample size, and the Rest of the world present relative high positive differences in the years after the M&A. On the other hand, M&As in Australia/New Zealand/Canada and Japan seem to realize the smallest differences, where Japan is the worst country group. The U.S. should be ranked in the middle. Summarizing the result from table 8, in general the results by country group resemble one another. Results are often statistically insignificant, but differences between actual and projected profits tend to be positive. The most profitable effects can be assigned to Continental Europe, the U.K., and the Rest of the World and the less profitable effects to Japan and Australia/New Zealand/Canada.

Table 9 reports the differences between actual and projected profits in the pre-crisis and crisis period. Here is the same approach used as in the event study. T-values and Wilcoxon Z-values

Table 9. Periodic firm performances in the 2003-2013 period, measured by profits.

Years	Difference in Mn $		% Positive		Years	Difference in Mn $		% Positive	
after the M&A	Pre-crisis period	Crisis period	Pre-crisis period	Crisis period	after the M&A	Pre-crisis period	Crisis period	Pre-crisis period	Crisis period
Total sample									
t+1	209.77***	-17.68	63.20%	56.10%					
N	250	376							
t+2	234.73***	31.76	63.20%	54.80%					
N	163	374							
t+3	197.75*	45.63	58.90%	56.20%					
N	95	349							
t+4	297.47	75.00	60.00%	55.60%					
N	45	322							
t+5		15.30		55.90%					
N		313							
United States					United Kingdom				
t+1	271.62***	-44.87	68.00%	56.70%	t+1	912.85	-205.50	66.70%	58.30%
N	97	157			*N*	3	12		
t+2	233.44***	75.32	66.10%	55.00%	t+2	-20.58	219.39	0.00%	66.70%
N	62	151			*N*	2	9		
t+3	103.81	34.57	69.70%	49.00%	t+3		82.32		66.70%
N	33	149			*N*		6		
t+4	130.40**	51.37	90.90%	49.60%	t+4		798.52		50.00%
N	11	133			*N*		4		
t+5		60.93		54.30%	t+5		1071.33		66.70%
N		127			*N*		3		
Australia/New Zealand/Canada					Continental Europe				
t+1	85.65	41.50	71.90%	56.30%	t+1	635.49**	-223.15	69.60%	46.40%
N	32	64			*N*	23	28		
t+2	-1.57	15.44	57.10%	56.70%	t+2	927.74***	-133.15	88.20%	44.40%
N	14	67			*N*	17	27		
t+3	-53.63	83.21	60.00%	65.10%	t+3	1,607.47***	14.62	90.90%	64.30%
N	5	63			*N*	11	28		
t+4	-849.44	23.68	0.00%	65.50%	t+4	2,392.57**	132.47	83.30%	60.70%
N	1	58			*N*	6	28		
t+5		-127.41		55.30%	t+5		100.23		52.00%
N		47			*N*		25		
Japan					Rest of world				
t+1	49.29	-28.56	49.30%	56.90%	t+1	115.16	142.45	70.00%	57.90%
N	75	58			*N*	20	57		
t+2	112.11	-60.32	53.60%	48.50%	t+2	150.03	93.92	75.00%	63.00%
N	56	66			*N*	12	54		
t+3	-8.38	-44.27	45.00%	53.70%	t+3	-286.39	195.35**	33.30%	66.70%
N	40	67			*N*	6	36		
t+4	-66.95	-67.09	41.70%	46.90%	t+4	17.41	258.63	66.70%	74.30%
N	24	64			*N*	3	35		
t+5		-51.85		52.50%	t+5		47.40		74.20%
N		80			*N*		31		

The Pre-crisis period includes the years 2003-2007, the Crisis period includes the years 2008-2013. Difference in Mn $ is the difference between actual and projected profits. A positive number therefore implies that the M&As increased profits, a negative number implies that the M&As decreased profits. N is the number of observations and t is the year of the M&A. P-values and Wilcoxon Z-values are used to determine the level of significance. P-values are used for samples > 30, Wilcoxon Z-values are used for samples < 30.

* Statistical significance at the 10% level.

** Statistical significance at the 5% level.

*** Statistical significance at the 1% level.

are used to measure the level of significance. The pre-crisis period contains observations in the years 2003-2007 and the crisis period contains observations in the years 2008-2013. The fifth post-merger years in the pre-crisis period are missing because the first M&As in the dataset take place in 2003. This means that the fifth post-merger years of those M&As are classified in the crisis period. Again the results in significance levels are mixed, but the total sample differences suggest a large difference between the pre-crisis and crisis period. M&As in the pre-crisis period seem to be largely profitable, while M&As in the crisis period seem to be less profitable and even negative in the first post-merger year (-17.68). Regarding the percentages positive returns, the numbers found, for both the pre-crisis and the crisis period, suggest that at least 54% of the differences are positive in all post-merger years. These results indicate that most M&As are profitable during a pre-crisis and a crisis period, but that the degree of profitability is less in a crisis period. Linking these results with the results from the event study, it strikes that the outcomes contradict each other. The results of the event study showed that M&A announcements were valued more positively during the crisis period than during the non-crisis period. However, the results of the profit analysis suggest that M&As are less profitable in a crisis period than in a non-crisis period.

Looking at the results per country group the same clear trend, where profitability is higher in the pre-crisis period than in the crisis period, is observable for the U.S. and Continental

Europe. However, the results are only statistical significant for the pre-crisis periods of those country groups. Regarding the other country groups, no distinct patterns can be found. Results are statistically insignificant and show mixed results for every country. Some differences are higher for the pre-crisis period and some others for the crisis-period.

5. Conclusion

"Sometimes your best investments are the ones you don't make" (Trump, 1987) was the first sentence of the introduction to illustrate the importance of making good investment decisions. This study finds that most M&A investments result in better firm performances, however, to underline Trump's statement, this study also finds that M&As do not always result in better firm performances. To be able to conclude this, two approaches are used to investigate the effects of M&As on firm performance. The first approach measured firm performance by the change in stock prices around the M&A announcement dates, while the second approach used the change in profits after an M&A as a measurement of firm performance. Both approaches focused on the differences between countries around the world and compared the non-crisis with the crisis period.

The first approach found that target shareholders perform better than acquirer shareholders around an M&A announcement date. Here target shareholders observe largely positive gains, while acquirer shareholders realize abnormal returns just above zero. A clear explanation for this stock price reaction is not found in the cross-sectional analysis, but from the comparison between time periods it can be concluded that M&A announcements in the crisis period provide higher stock returns than M&A announcements in the non-crisis period. This is a trend observed for both acquiring and target firms and applies to almost all country groups in the sample. A clear explanation for this trend is not found, but maybe it is due to the better consideration of engaging in M&A activity in a crisis-period or the M&A announcements are valued more positively in a crisis period as it can result in a turnaround for struggling firms.

The profit analysis in the second approach showed positive differences between the actual and projected profits of firms. This means that M&As are in general profitable in at least the first five years after the M&A. In contrast to the effect of M&As on stock prices, the change in profits due to an M&A seem to be more positive in a non-crisis period than in a crisis period.

The findings specified per country show no major differences between countries for both approaches. This means that shareholders of acquiring and target firms react mostly positive on M&A announcements in all countries and that profits, although the mixed results in significance levels, seem to increase in almost all country groups after an M&A. The most interesting finding here is how similar the M&A effects look across the different countries, which confirms the conclusions of Gugler et al. (2002). Making a ranking in country groups is not possible because there is no country group that shows structural the most positive or negative effects. Most important in the magnitude of the effects are probably the firm and M&A specific characteristics which need to be re-examined each time to make a realistic expectation of the M&A effect. The literature showed for example that M&A specific characteristics, like deal attitude and payment method, and firm specific characteristics, like firm size and business relatedness, could influence the M&A effect. Due to these characteristics it is not possible to present a clear M&A effect which applies to all M&As. Therefore, to make a realistic expectation of the magnitude of the M&A effect, it is important to look at the characteristics for each M&A separately.

6. Limitations and further research

This thesis studied the M&A effects from several perspectives. However, there are some limitations to this study that might influence the results and which could be taken into account in additional studies. First of all, a comparison between the pre-crisis and crisis period is made, where the pre-crisis period contains the years 2002-2007 and the crisis period the years 2008-2013. However, different economic situations across countries are not taken into account, while the duration of the period in which countries are in crisis could differ from the assumed 2008-2013 period. Secondly, mostly due to the sample requirements, in some cases the sample sizes became small when the results were specified per country and year/period, which made it not always possible to find clear results. Thirdly, the cross-sectional analysis contained a limited number of variables. Adding more variables, like firm size and deal atmosphere, could give more significant results and a better explanation of the CARs. Fourthly, to measure firm performance by profits the formula of Gugler et al. (2002) is used. Besides this formula, it is also possible to use other methods, like profit rates, which may result in more significant results. Finally, country groups are classified in the same way as Gugler et al. (2002) did. When the country groups are divided in a different way, a change in conclusions is not expected, but maybe it leads to outcomes that allow for a clear ranking of countries.

In addition to this study, further research could use a different dataset to increase the limited number of observations, do robustness checks using different event windows and estimation periods, look at specific industries in different countries or re-examine the results by comparing the M&A effects during other non-crisis and crisis periods. Furthermore, this study found that M&A announcements during the crisis provide higher stock returns than M&A

announcements in the non-crisis period. Further research could focus on this finding and try to find out why the returns in crisis periods are higher.

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